Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 35

Bitcoin Consensus

Estimated Reading Time: 15–20 minutes

Pre-Reading Material: Bitcoin Consensus

Estimated Reading Time: 5 minutes

Introduction

Bitcoin consensus is the process through which computers in the Bitcoin network agree on a common, valid record of transactions without depending on a central authority. It ensures that participants follow the same rules when verifying transactions and maintaining the blockchain.

Proof of Work (PoW)

Bitcoin uses Proof of Work (PoW) as its consensus mechanism. In this approach, miners compete to find a valid block by performing computational work. The winning miner can broadcast the block to the network, where other nodes independently verify its validity.

Mining requires considerable computational resources, while checking a proposed block is comparatively easy. This difference helps protect the blockchain against unauthorized changes.

Working of Bitcoin Consensus

The Bitcoin consensus process involves the following steps:

- Transaction broadcasting: Users broadcast their transactions to the Bitcoin network.
- Transaction verification: Nodes check whether transactions follow Bitcoin's rules.
- Block creation: Miners collect valid transactions into candidate blocks.
- Proof of Work: Miners perform computational work to find a valid block.
- Block validation: Other nodes independently verify the proposed block.

Campus : Prestige Vihar, Scheme No. 74-C, Sector D, Vijay Nagar, Indore -452010 (M.P.) India
Phone : 0731-4013333, Fax : 0731-4013329 E-mail : info@piemr.edu.in, Web. www.piemr.edu.in

- Chain selection: Nodes follow the valid blockchain with the greatest accumulated proof of work.

This process helps the distributed network maintain a consistent transaction history.

Importance of Bitcoin Consensus

Bitcoin consensus provides decentralization, transaction integrity, and protection against double spending. It makes it difficult to alter previously confirmed transactions because changing an earlier block requires overcoming the proof of work supporting the subsequent chain.

However, confirmations do not provide an absolute guarantee against transaction reversal. Additional confirmations generally increase confidence that a transaction will remain in the blockchain.

Key Takeaway

Bitcoin consensus combines Proof of Work, independent transaction validation, and blockchain chain-selection rules to enable computers worldwide to maintain a shared transaction ledger without a central authority.

Pre-Reading Questions

1. What is consensus in Bitcoin?
2. Which consensus mechanism does Bitcoin use?
3. What is the role of miners?
4. Why do nodes validate blocks independently?
5. How does Bitcoin consensus help prevent double spending?

Campus : Prestige Vihar, Scheme No. 74-C, Sector D, Vijay Nagar, Indore -452010 (M.P.) India
Phone : 0731-4013333, Fax : 0731-4013329 E-mail : info@piemr.edu.in, Web. www.piemr.edu.in